

Departement Elektriese, Elektroniese en
Rekenaar-Ingenieurswese

Department of Electrical, Electronic and
Computer Engineering

Semestertoets 3

Kopiereg voorbehou

Semester Test 3

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Module EBN122
Elektrisiteit en Elektronika
November 2016

Module EBN122
Electricity and Electronics
November 2016



UNIVERSITEIT VAN PRETORIA
UNIVERSITY OF PRETORIA
YUNIBESITHI YA PRETORIA

Exam information:

Eksamensinligting:

Maximum marks: Maksimum punte:	45	Full marks: Volpunte:	46
Duration of paper: Duur van vraestel:	90 minutes 90 minute	Open / closed book: Oopboek / toeboek:	Closed Toe
Total number of pages (including this page): Totale aantal bladsye (hierdie blad ingesluit):	13		

IMPORTANT - BELANGRIK

- The examination regulations of the University of Pretoria apply.*
Die eksamenregulasies van die Universiteit van Pretoria geld.
- All questions must be answered on the ANSWER SHEET.*
Alle vrae moet op die ANTWOORDBLAD beantwoord word.
- Students may do their own rough calculations on the question paper – the question paper will not be taken in.*
Studente mag hul eie rowwe berekeninge op die vraestel doen – die vraestel word nie ingeneem nie.
- Answers must include units!*
Antwoorde moet eenhede insluit!
- Final answers must be written as real numbers, and not as fractions.*
Finale antwoorde moet as reële getalle geskryf word, en nie as breuke nie.

Dosent(e): Lecturer(s):	1. D le Roux
	2. R dos Santos
	3. J Joubert

VRAAG/QUESTION 1 [4]

Given: $V_3 = 8 \text{ V}$ and $V_2 = 6 \text{ V}$.

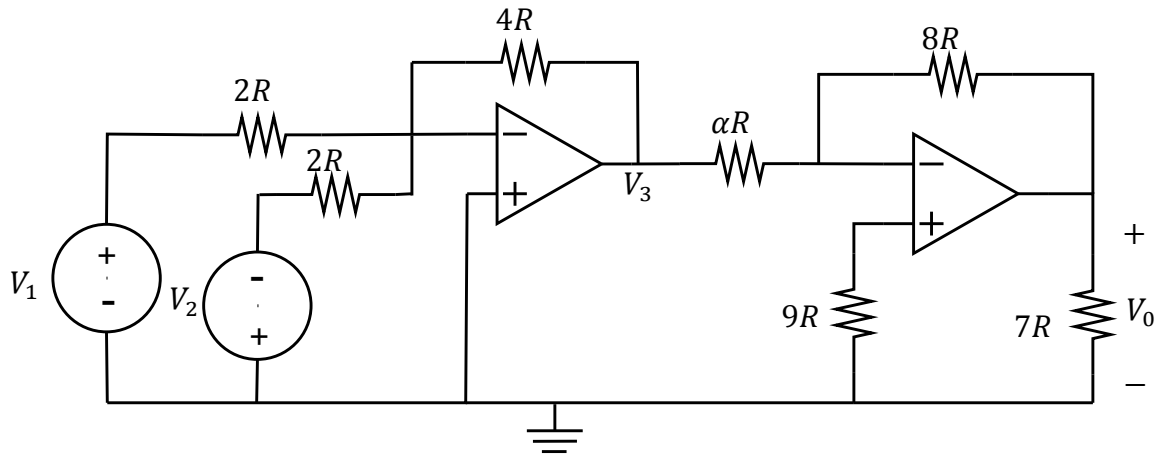
a) Calculate V_1 .

b) Find the value of α for which $V_0 = -16 \text{ V}$.

Gegee: $V_3 = 8 \text{ V}$ en $V_2 = 6 \text{ V}$.

a) Bereken V_1 .

b) Vind die waarde vir α sodat $V_0 = -16 \text{ V}$.



VRAAG/QUESTION 2 [10]

Question 2.1 [2]

The voltage across a 2 F capacitor is given as $v(t) = 8 - 3e^{-5t}V$. The current through the capacitor at $t = 0$ is given as 30 A. Determine the current at $t = 2$ s.

Vraag 2.1 [2]

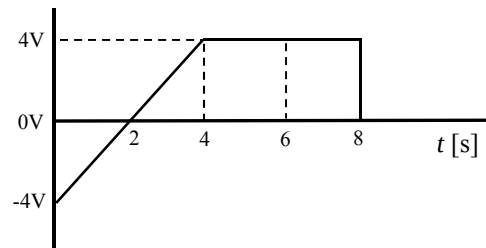
Die spanning oor 'n 2 F kapasitor is $v(t) = 8 - 3e^{-5t}V$. Die stroom deur die kapasitor by $t = 0$ s is 30 A. Bepaal die stroom by $t = 2$ s.

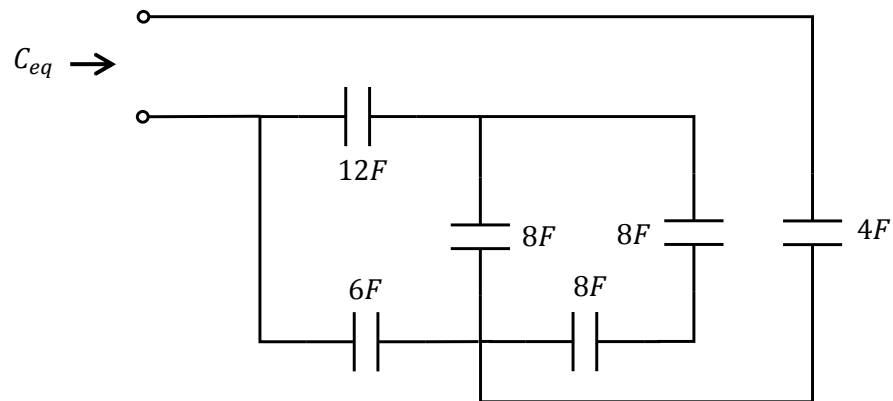
Question 2.2 [2]

The graph below shows the voltage across a 10 H inductor. Assuming the initial current through the inductor is 4 A, what is the current between 0 and 6 seconds?

Vraag 2.2 [2]

Onderstaande grafiek dui die spanning oor 'n 10 H induktor aan. Indien die aanvanklike stroom deur die induktor 4 A is, wat is die stroom tussen 0 en 6 sekondes?



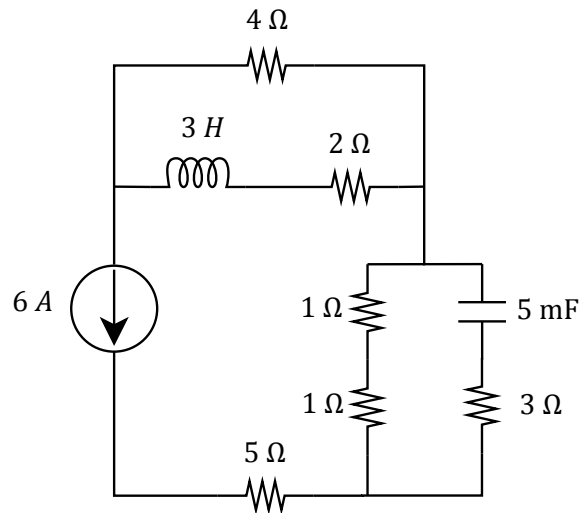
Question 2.3 [2]Calculate C_{eq} .**Vraag 2.3 [2]**Bepaal C_{eq} .

Question 2.4 [4]

Calculate the energy stored by the capacitor (w_C) and inductor (w_L). Assume direct current conditions.

Vraag 2.4 [4]

Bepaal die energie gestor in die kapasitor (w_C) en die inductor (w_L). Aanvaar gelykstroom toestande.



VRAAG/QUESTION 3 [16]

Question 3.1 [6]

- a) Write $v(t)$ as a phasor in **rectangular form**:
 $v(t) = 10\cos(\omega t - 90^\circ) - 30\sin(\omega t + 45^\circ)$ V
- b) Determine the load impedance if
 $v_L(t) = 300\cos(5000\pi t)$ V, and
 $i_L(t) = 6\sin(5000\pi t)$ A.
Assume the current is flowing into the + of the reference voltage across the load. Write the answer in **polar form**.
- c) A 20Ω resistor and a $2\mu F$ capacitor are connected in parallel. A voltage source $v(t) = 500\cos(5000t + 20^\circ)$ V is connected across the parallel combination. Determine the current $i_C(t)$ that flows through the capacitor. Assume the current is flowing from the source into the + of the reference voltage defined across the load impedance.

Vraag 3.1 [6]

- a) Skryf $v(t)$ as 'n fasor in **reghoekige vorm**:
 $v(t) = 10\cos(\omega t - 90^\circ) - 30\sin(\omega t + 45^\circ)$ V
- b) Bepaal die las-impedansie as
 $v_L(t) = 300\cos(5000\pi t)$ V, en
 $i_L(t) = 6\sin(5000\pi t)$ A.
Aanvaar dat die stroom by die + van die verwysingspanning oor die las invloei. Skryf die antwoord in **polêre vorm**.
- c) 'n 20Ω weerstand en 'n $2\mu F$ kapasitor is in parallel geskakel. 'n Spanningsbron $v(t) = 500\cos(5000t + 20^\circ)$ V word oor die parallel-kombinasie aangelê. Bereken die stroom $i_C(t)$ wat deur die kapasitor vloei. Aanvaar dat die stroom vanaf die bron by die + van die verwysingspanning gedefinieer oor die las-impedansie invloei.

Question 3.2 [4]

a) Simplify and write the answer in **rectangular form**:

$$\bar{I}_1 = \frac{(3 + j4) + 11\angle 90^\circ}{(2\angle 45^\circ)^2 + (3 - j4)}$$

b) Determine $\bar{V}_2$ in **polar form**:

$$\begin{bmatrix} 2 + j4 & -j2 \\ -j2 & 2 \end{bmatrix} \begin{bmatrix} \bar{V}_1 \\ \bar{V}_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 2\angle 90^\circ \end{bmatrix}$$

Vraag 3.2 [4]

a) Vereenvoudig en skryf die antwoord in **reghoekige vorm**:

$$\bar{I}_1 = \frac{(3 + j4) + 11\angle 90^\circ}{(2\angle 45^\circ)^2 + (3 - j4)}$$

b) Bepaal $\bar{V}_2$ in **polêre vorm**:

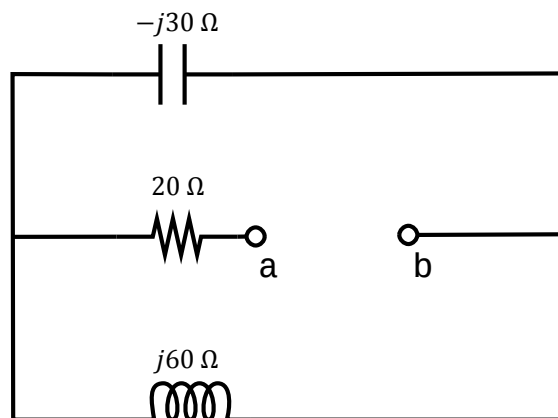
$$\begin{bmatrix} 2 + j4 & -j2 \\ -j2 & 2 \end{bmatrix} \begin{bmatrix} \bar{V}_1 \\ \bar{V}_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 2\angle 90^\circ \end{bmatrix}$$

Question 3.3 [2]

Determine the total impedance $\bar{Z}_{ab}$ between terminals a and b . Write the answer in **rectangular form**.

Vraag 3.3 [2]

Bepaal die totale impedansie $\bar{Z}_{ab}$ tussen terminale a en b . Skryf die antwoord in **reghoekige vorm**.

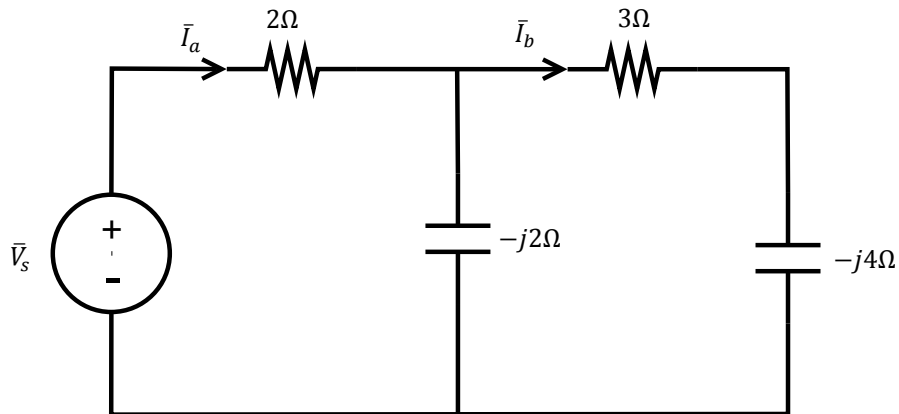


Question 3.4 [4]

- a) Determine $\bar{I}_a$ if $\bar{I}_b = 1 \angle -90^\circ \text{ A}$. Write the answer in **rectangular form**.
- b) Determine $\bar{I}_a$ if $\bar{V}_s = 2 \angle 60^\circ \text{ V}$. Write the answer in **polar form**.

Vraag 3.4 [4]

- a) Bepaal $\bar{I}_a$ as $\bar{I}_b = 1 \angle -90^\circ \text{ A}$. Skryf die antwoord in **reghoekige vorm**.
- b) Bepaal $\bar{I}_a$ as $\bar{V}_s = 2 \angle 60^\circ \text{ V}$. Skryf die antwoord in **polêre vorm**.



VRAAG/QUESTION 4 [16]

Question 4.1 [4]

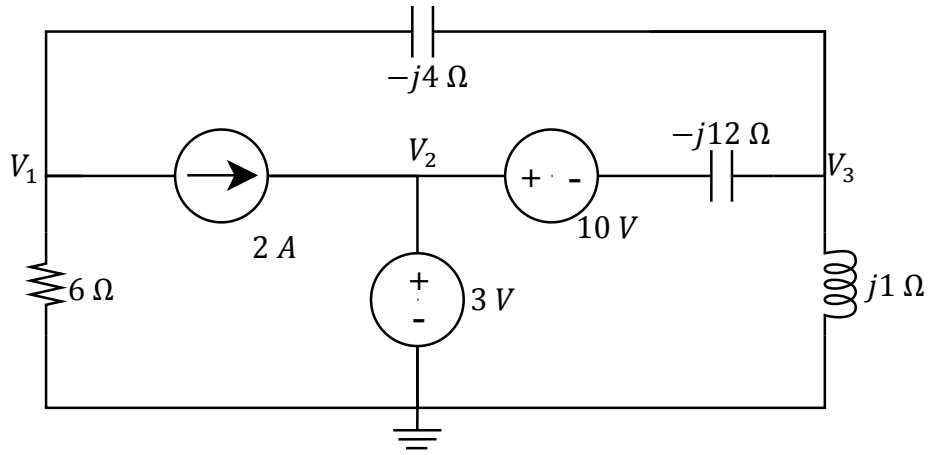
Use nodal analysis to set up two equations of the following form:

$$(a_1 + ja_2)V_1 + (b_1 + jb_2)V_3 = -24$$
$$cV_1 + dV_3 = 7$$

Vraag 4.1 [4]

Gebruik node analyse om twee vergelykings van die volgende vorm op te stel:

$$(a_1 + ja_2)V_1 + (b_1 + jb_2)V_3 = -24$$
$$cV_1 + dV_3 = 7$$



Question 4.2 [4]

Use mesh analysis to set up two equations of the following form:

$$(e_1 + je_2)\bar{I}_1 + (f_1 + jf_2)\bar{I}_2 = -3$$

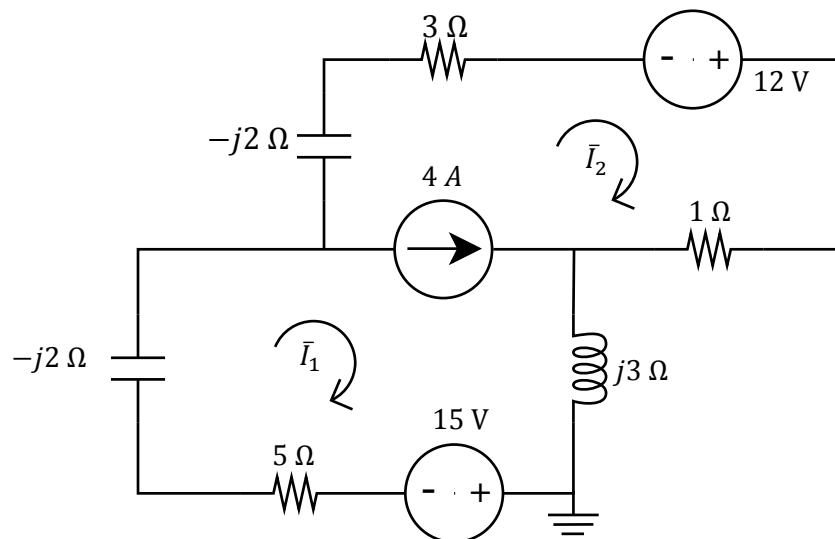
$$g\bar{I}_1 + h\bar{I}_2 = 4$$

Vraag 4.2 [4]

Gebruik lusanalyse om twee vergelykings van die volgende vorm op te stel:

$$(e_1 + je_2)\bar{I}_2 + (f_1 + jf_2)\bar{I}_3 = -3$$

$$g\bar{I}_1 + h\bar{I}_2 = 4$$



Question 4.3 [4]

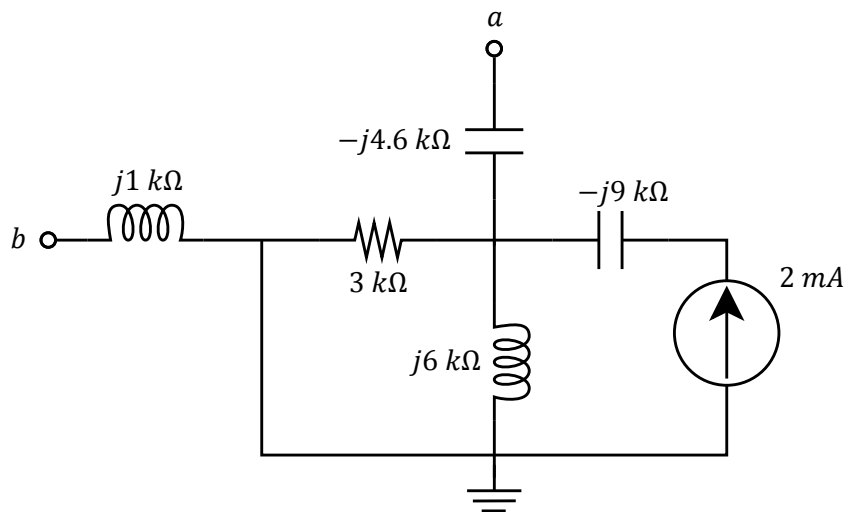
Find the Thevenin equivalent of the circuit with respect to terminals a - b .

- a) $\bar{V}_{TH}$ in **rectangular** form
- b) Z_{TH} in **rectangular** form

Vraag 4.3 [4]

Vind die Thevenin ekwivalent met verwysing tot terminale.

- a) $\bar{V}_{TH}$ in **reghoekige** vorm
- b) Z_{TH} in **reghoekige** vorm.



Question 4.4 [4]

Find the Norton equivalent of the circuit with respect to terminals a - b .

a) $\bar{I}_N$ in **rectangular** form

b) Z_N in **polar** form

Vraag 4.4 [4]

Vind die Norton ekwivalent met verwysing tot terminale a - b .

a) $\bar{I}_N$ in **reghoekige** vorm

b) Z_N in **polêre** vorm.

